

AI-Integrated Instruction-Based Processing System

Objective

To build a web-based AI assistant that responds to user queries based only on the information provided by the Admin and Users, simulating a ChatGPT-like experience. This application is divided into two main modules:

- Admin Section – To manage and train the AI model
- User Section – To interact with the AI using chat and upload documents for additional training

Key Features

1. Admin Panel

- Upload documents (PDF, DOCX, TXT, CSV, etc.)
- Feeding informations
- Train the AI model using these documents
- Tag and categorize uploaded content
- View history of uploaded files
- Trigger model re-training manually if needed
- Manage user-uploaded content (approve/deny)
- Dashboard with usage insights (optional)

2. User Panel

- Chat with the AI model (chat interface)
- Upload documents for training (requires admin approval or auto-training)
- View previous conversations
- Authentication system (Login/Registration)
- Ask questions and give instructions based on fed knowledge

AI Model Behavior

- The AI has no access to external information (internet or default GPT knowledge).
- It is trained only on the uploaded datasets (vector database / embedding-based search).
- Any data uploaded and accepted becomes a permanent part of the AI's knowledge base.
- Embeddings are stored in a vector database (e.g., Pinecone, Weaviate, Qdrant, or FAISS).
- AI retrieves relevant information using semantic search and answers based on that context.

Tech Stack (Suggested)

Layer

Technology

Frontend	React / Vue / Next.js
Backend	Laravel / Node.js / Django
AI Model	OpenAI GPT-4 / GPT-3.5 via API or open-source like LLaMA2
File Parsing	LangChain / Unstructured / PyMuPDF / Tika
Vector Store	Pinecone / FAISS / Weaviate
Authentication	JWT / Laravel Sanctum / Firebase Auth
File Storage	AWS S3 / Firebase Storage / Local Server
Database	PostgreSQL / MySQL / MongoDB

Modules Breakdown

A. Admin Features

Feature	Description
File Upload	Upload PDF, Word, and text documents
Document Management	Categorize, tag, delete, or update documents
Training Monitor	Trigger or monitor training and embedding creation
AI Testing Console	Admin can test AI responses using a sample chat
User Document Approval	Approve/deny user-uploaded documents before training

B. User Features

Feature	Description
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Chat Interface	Talk to AI based on trained knowledge
Upload Document	Upload own files to extend knowledge (admin-controlled)
Document Status	Check if uploaded documents are accepted and trained
Session History	View chat history for their sessions
Login/Sign-up	Authentication for secured access

Data Persistence & Security

- All embeddings and document knowledge will be persisted in vector database, not lost after sessions.
 - User documents can be sandboxed or globally trained based on business logic.
 - Proper access control and encryption for file storage.
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Workflow

Admin Flow:

1. Login → Upload Documents / feeding informations → AI Vector Embedding → Model Ready
2. Some basic settings

User Flow:

1. Register/Login → Upload Document (optional) → Ask Questions → Receive AI Response

 Use Cases

- Internal knowledge assistant for a company
- AI assistant for custom documentation
- Private knowledge Q&A bot (e.g., for law firms, medical orgs, school syllabus)